Zero-Shot OOD Detection with CLIP: Reproducing MCM, L-MCM and GL-MCM

COME421/1 Introduction to Computer Vision · Midterm project · Progress report
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Article: Miyai, A., Yang, J., Zhang, J., Ming, Y., Lin, Y., Yu, Q., Irie, G., Joty, S., Li, Y., Li, H., Liu, Z., Aizawa, K., "GL-MCM: Global and Local Maximum Concept Matching for Zero-Shot Out-of-Distribution Detection", International Journal of Computer Vision (IJCV), 2025. DOI 10.1007/s11263-025-02356-z.

Project repository: https://github.com/Yusuflora/cv-gl-mcm-project (fork of the authors' code, adapted to run on Apple GPU).

1. Aim of the project

Out-of-distribution (OOD) detection asks a classifier to recognise inputs that do not belong to any of its known classes. The article proposes **GL-MCM**, a training-free, zero-shot method built on CLIP: besides the *global* image–text matching score used by the earlier MCM baseline, it adds a *local* score computed from image regions, so that small ID objects inside a cluttered multi-object scene are not missed. Our midterm goal is to implement the methods compared in the article, run them on the article's datasets with the same metrics (FPR95, AUROC) and present the results in tables and plots.

The two properties we study (with / without) are the **global** matching score (MCM) and the **local** matching score (L-MCM), and what happens when both are combined (GL-MCM).

2. Methods implemented so far

Let s_i be the cosine similarity between an image feature and the text feature of ID class name i (prompt "a photo of a <class>"), and τ the temperature.

Method	Score (higher = more in-distribution)	Meaning
MCM	$S_{MCM}(x) = \max_i \exp(s_i/\tau) / \Sigma_j \exp(s_j/\tau)$, using the **global** image feature	Baseline: how strongly the whole image matches its best ID class.
L-MCM	$S_{L\text{-}MCM}(x)$ = max over image regions r and classes i of the softmax score computed on **local** features	Looks for any region that matches an ID class.
GL-MCM	$S_{GL\text{-}MCM}(x) = S_{MCM}(x) + \lambda \cdot S_{L\text{-}MCM}(x)$	Combines global and local evidence; λ weights the local term.

An image is flagged as OOD when its score is below a threshold. **FPR95** is the fraction of OOD images wrongly accepted when 95% of ID images are correctly accepted (lower is better); **AUROC** and **AUPR** are threshold-free ranking metrics (higher is better).

3. Project flow diagram

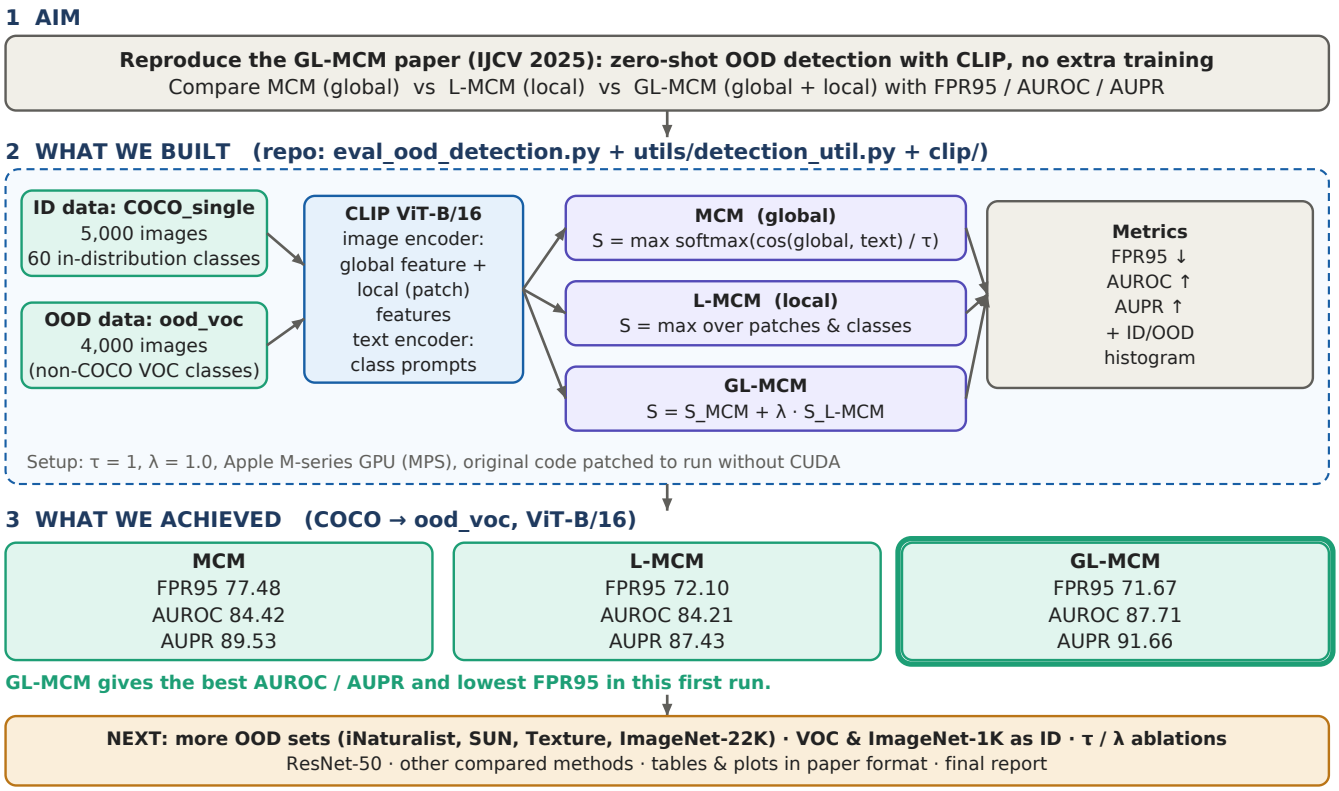


Figure 1 – Aim, pipeline (as implemented in the repository) and first results. Green = done, amber = still to do.

4. What we did

Step	Description	Status
1	Studied the article and the instructor's midterm requirements; mapped them to the authors' open-source repository (AtsuMiyai/GL-MCM).	Done
2	Created a GitHub repository to share the code and progress.	Done
3	Set up Python 3.12 environment with PyTorch (Apple MPS). Patched the code, which assumed an NVIDIA GPU (CUDA): removed CUDA assertions and moved tensors to the Apple GPU; added an **--out_list** option to select OOD sets.	Done
4	Downloaded and extracted the authors' dataset package (ID_COCO_single, ID_COCO_multi, ID_VOC_single, OOD_COCO, OOD_VOC).	Done
5	Ran MCM, L-MCM and GL-MCM (CLIP ViT-B/16, τ = 1, λ = 1.0) with COCO_single as ID (5,000 images, 60 classes) and ood_voc as OOD (4,000 images). Each run takes about 7 minutes on a MacBook Air.	Done
6	Collected FPR95 / AUROC / AUPR in CSV files and ID-vs-OOD score histograms for each method.	Done
7	Remaining OOD sets, other ID sets, ablations, remaining compared methods, final report.	To do

5. What we achieved – first results

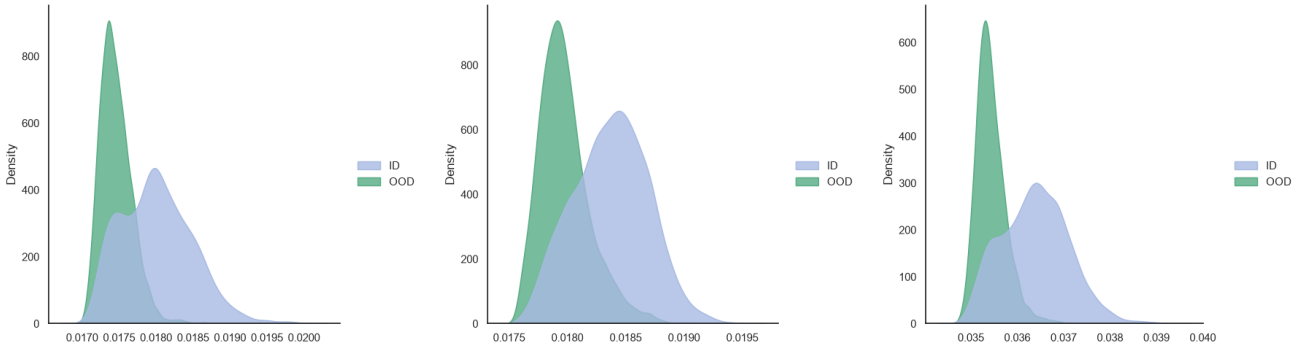
Setting: ID = COCO_single (60 classes, 5,000 images), OOD = ood_voc (4,000 images), CLIP ViT-B/16, $\tau = 1.0$, $\lambda = 1.0$ for GL-MCM. Single run, no averaging over several OOD datasets yet.

Method	FPR95 ↓	AUROC ↑	AUPR ↑
MCM (global)	77.48	84.42	89.53
L-MCM (local)	72.10	84.21	87.43
GL-MCM (global + local)	**71.67**	**87.71**	**91.66**

Table 1 – COCO → ood_voc. Values are percentages.

Observations

- **Local information alone lowers FPR95** (77.48 → 72.10) but does not improve ranking quality: AUROC and AUPR are slightly lower than MCM (84.21 vs 84.42; 87.43 vs 89.53).
- **Combining both works best**: GL-MCM improves over MCM by −5.81 points FPR95, +3.29 points AUROC and +2.13 points AUPR, and over L-MCM in every metric. This matches the article's claim that global and local scores are complementary.
- The histograms show why: with GL-MCM the OOD scores (green) form a narrow peak at low values while the ID scores (blue) are shifted to the right, so the overlap region is smaller than for MCM.



(a) MCM (b) L-MCM (c) GL-MCM

Figure 2 – Distribution of OOD scores for ID (blue, COCO) and OOD (green, ood_voc) images. Note that the x-axis ranges differ between the three plots (MCM ≈ 0.017–0.020, GL-MCM ≈ 0.035–0.040) because GL-MCM sums two scores.

Score statistics (from the evaluation logs)

Method	ID mean score (n = 5,000)	OOD mean score (n = 4,000)
MCM	0.018006	0.017458
L-MCM	0.018373	0.017978
GL-MCM	0.036379	0.035435

Scores are stored with a negative sign in the evaluation code; the table shows magnitudes. Because $\tau = 1$ and there are 60 classes, softmax scores stay close to $1/60 \approx 0.0167$, which is why the absolute values look small – only their ranking matters for FPR95 / AUROC.

6. Limitations of the current results

- Only **one** OOD dataset (ood_voc) and one ID dataset (COCO_single) so far; the article averages over several OOD sets, so these numbers are **not yet comparable** with the article's tables.
- Single run per method with the default $\lambda = 1.0$; no τ or λ study yet.
- Only the three CLIP-based zero-shot scores are implemented. The article also compares few-shot (CoOp, LoCoOp) and supervised baselines (MSP, Energy, ODIN, ViM, KNN, NPOS) and other methods; their scope for this midterm is being confirmed with the instructor.
- The code was patched for Apple GPU (MPS); small numerical differences from the authors' CUDA runs are possible.

7. Plan for the remaining work (before 19 Nov 2026)

#	Task	Midterm requirement addressed
1	Download the missing OOD sets (iNaturalist, SUN, Texture, ImageNet-22K, Places) and ImageNet-1K validation set.	Use the article's datasets
2	Run MCM / L-MCM / GL-MCM on COCO_single, VOC_single and ImageNet-1K with all their OOD sets (optionally COCO_multi).	Implement the compared methods; same metrics
3	Ablations: temperature τ (article Table 3), local weight λ (Fig. 3), ID/OOD histograms (Fig. 4), ResNet-50 backbone (Table 5).	Tables and plots
4	Implement / run further compared methods, depending on the instructor's answer on scope (trained baselines need extra training).	Methods compared in the article
5	Produce results tables and plots in the article's format and compare with the article's reported numbers.	Tables, plots
6	Write the report: explanation of each code section and output; zip and upload via STIX before 19 Nov 2026; prepare the presentation (19 Nov).	Report and submission

8. Reproducibility

Command used for each score (S ∈ {MCM, L-MCM, GL-MCM}):

```
PYTORCH_ENABLE_MPS_FALLBACK=1 python eval_ood_detection.py --in_dataset COCO_single --name test --CLIP_ckpt ViT-B/16 --score S --root-dir ./datasets --num_ood_sumple 500 --out_list ood_voc
```

Outputs per run: **results/COCO_single/<S>/CLIP_ViT-B_16_T_1_ID_test/** contains test.csv (metrics), ood_eval_info.log (score statistics) and the histogram PNG.